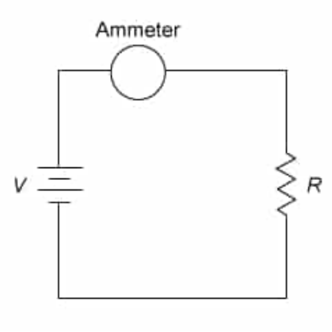
A battery of unknown voltage V is connected to a resistor R and an ammeter as shown.

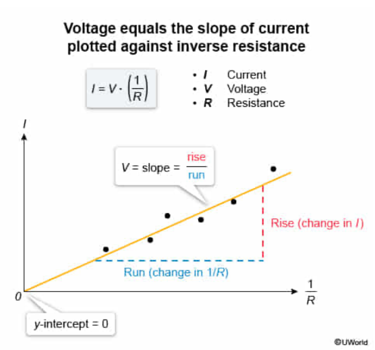


Researchers change the value of R and record the reading of the ammeter each time. The value of V can best be determined from which of the following?

- A. The slope of a graph of current as a function of R [27%]
- B. The slope of a graph of current as a function of $1/R$ [50%]
- C. The *y*-intercept of a graph of current as a function of R [14%]
- D. The *y*-intercept of a graph of current as a function of $1/R$ [7%]

Correct
50% answered correctly

Explanation:



Ohm's law states that the current I between two points in a closed circuit is directly proportional to the **voltage** V between those two points and inversely proportional to the resistance R:

$$I = \frac{V}{R}$$

In this question, researchers are trying to determine the unknown V of a battery by using an ammeter to measure the current when R changes. If the data for the current are plotted on the *y*-axis of a graph (as the *y*-variable) and the data for the *inverse* of resistance $\frac{1}{R}$ are plotted on the *x*-axis (as the *x*-variable), then Ohm's law can be expressed as a **linear function** in slope (m)-intercept (b) form:

$$y = m \cdot x + b$$
$$\Downarrow$$
$$I = V \cdot \left(\frac{1}{R}\right) + 0$$

When Ohm's law is expressed in this way, the slope of the line is equal to V and the *y*-intercept is zero. Therefore, the value of V can be determined from the slope of the graph of current as a function of $\frac{1}{R}$.

(Choice A) The graph of current as a function of resistance would yield an inverse relationship curve rather than a straight line with a slope that can be calculated as function of R.

(Choices C and D) The *y*-intercept of a graph of I versus R (or I versus $\frac{1}{R}$) would give the value of the current only when R (or $\frac{1}{R}$) equals zero. Therefore, the *y*-intercept does not yield any information about the voltage of the battery.

Educational objective:
Ohm's law can be expressed as a linear relationship between the current and the inverse of resistance, with voltage equal to the slope of the line.

Time spent:0 sec
Subject:
Physics

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System:
Electrostatics and Circuits